

Robert Ford

CONTACT

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PROFESSIONAL EXPERIENCE

Alpha Coding | 2021 – 2022

Alpha Coding Private Tutors in Toronto

- Tutored children (10–17) in Python fundamentals, and advanced concepts
 - Wrote up lesson plans and tracked learning progression
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EDUCATION

University of Toronto | 2021 – 2025

Bachelor Of Science

- Majored in Statistics
 - Minored in Computer Science and History
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PROJECTS

Information Sharing in Bayesian Neural Networks | 2024

In Bayesian Neural Networks, the model parameters are treated as random variables with probability distributions rather than fixed point estimates. In this project, I used Bayesian hierarchical modelling to share statistical strength across units, which helps regularize the network and stabilize activations, reducing issues like dead units and activation/gradient blow-up.

Youtube Agent | 2025

Using MCP and other industry-standard tooling, I built an automated YouTube Shorts LLM-driven agent that generates short-form videos about historical events and publishes them end-to-end (creation → rendering → upload). On the backend, I implemented a data pipeline that pulls YouTube Analytics and performance signals (e.g., watch time, likes) and applies statistical optimization and experimentation techniques to iteratively improve content quality and engagement over time.

Preventing Model Collapse | 2025

Model collapse refers to the progressive degeneration of generative models when they are repeatedly trained on their own synthetic outputs, which leads to catastrophic loss of diversity in the outputs. To mitigate this, I used a bi-level optimization setup where: an outer-loop foundation model is optimized using error signals from inner-loop models trained on synthetic data. This trains the foundation model to produce synthetic datasets that reduce model degradation and makes downstream training more robust to model collapse.